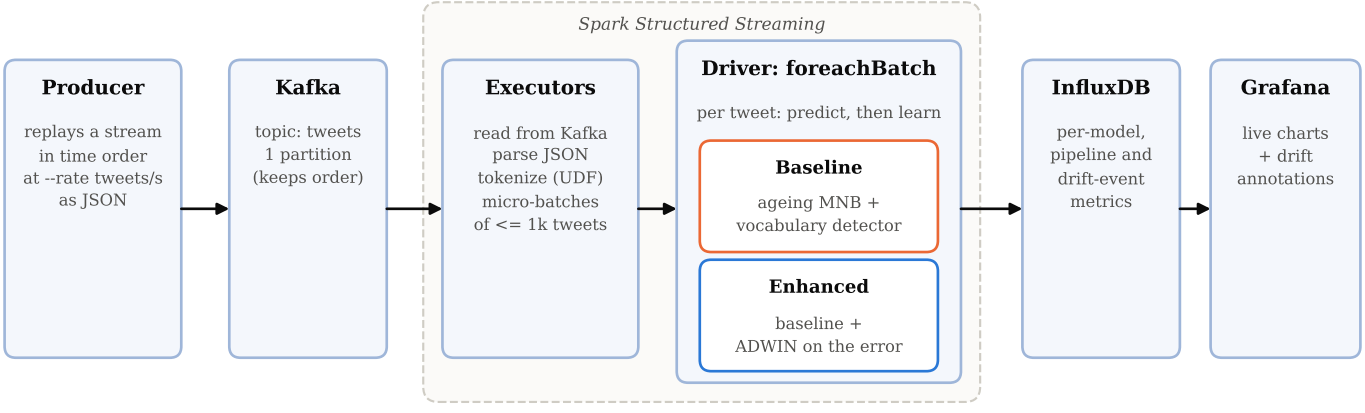
Spark Structured Streaming
Producer
replays a stream
in time order
at --rate tweets/s
as JSON
Kafka
topic: tweets
1 partition
(keeps order)
Executors
read from Kafka
parse JSON
tokenize (UDF)
micro-batches
of <= 1k tweets
Driver: foreachBatch
per tweet: predict, then learn
Baseline
ageing MNB +
vocabulary detector
Enhanced
baseline +
ADWIN on the error
InfluxDB
per-model,
pipeline and
drift-event
metrics
Grafana
live charts
+ drift
annotations